Bangladesh Army University of Engineering &Technology (BAUET)
Department of Computer Science and Engineering (CSE)
Course Title: Machine Learning Course Code: CSE-4115
4th Year 1st Semester Batch: 17th Summer-26
Marks: 15

In this project, you are required to apply machine learning techniques to solve a real-world problem using a publicly available dataset. You may choose one dataset from the following options: Titanic Survival Prediction, House Price Prediction, Student Performance Prediction, or Heart Disease Prediction. The datasets are available on Kaggle and can be accessed through their respective links.

i. https://www.kaggle.com/datasets/yasserh/titanic-dataset
ii. https://www.kaggle.com/datasets/harlfoxem/housesalesprediction
iii. https://www.kaggle.com/datasets/spscientist/students-performance-in-exams
iv. https://www.kaggle.com/datasets/fedesoriano/heart-failure-prediction

You will begin by loading the dataset using Python and performing data preprocessing, which includes handling missing values, encoding categorical variables, and applying feature scaling techniques where necessary. After preprocessing, you will conduct exploratory data analysis by visualizing the data using graphs such as histograms, heatmaps, and pair plots, and then provide meaningful insights based on your observations.

Next, you will split the dataset into training and testing sets and develop at least two machine learning models suitable for the problem, such as Naïve Bias, Logistic Regression, Decision Tree, K-Nearest Neighbors, or Random Forest. After building the models, you will evaluate their performance using appropriate metrics. For classification problems, you should use accuracy, precision, recall, F1-score, and confusion matrix, while for regression problems, you should use MAE, MSE, RMSE, and R^2 score.

You are also required to improve at least one model by applying techniques such as hyperparameter tuning and cross-validation. After evaluation and improvement, you will analyze the results to determine which model performed best and explain the reasons behind its performance, including identifying the most important features influencing the predictions.

Finally, you will conclude the project by summarizing your findings, discussing any limitations of your model, and suggesting possible future improvements. As an optional extension, you may implement advanced models such as Support Vector Machines or XGBoost, perform feature engineering.

The final submission should include a well-documented Jupyter Notebook containing the code and outputs, along with a report of approximately explaining your methodology, analysis, and results.